EC2 Instance Storage & EFS - AWS SAA-C03 Study Notes

1. EBS (Elastic Block Store) Overview

- **Definition:** A durable, block-level storage device that you can attach to a single EC2 instance. It acts like a virtual hard drive (network drive).
- **Availability Zone (AZ) Locked:** An EBS volume is bound to a specific AZ. To move a volume to another AZ, you must take a snapshot and restore it in the new AZ.
- **Persistence:**
  - By default, the **Root Volume** (where the OS is installed) is set to **Delete on Termination**.
  - Additional (data) volumes are usually set to **Keep** by default.
  - You can change this behavior via the "DeleteOnTermination" attribute.

2. EBS Volume Types

You must know the difference between SSD (performance) and HDD (throughput) for the exam.

Solid State Drives (SSD) - Optimized for transactional workloads (IOPS)

Type	Name	Use Case	Key Features
General Purpose	**gp3**	**Default choice**. Boot volumes, virtual desktops, dev/test.	**Independent** scaling of IOPS and Throughput. Baseline 3,000 IOPS & 125 MiB/s. Cheaper than gp2.
General Purpose	**gp2**	Legacy choice.	IOPS linked to volume size (3 IOPS per GB). Max 16,000 IOPS. Uses "Burst Balance" for small volumes.
Provisioned IOPS	**io2 Block Express**	**Mission-critical**, sub-millisecond latency.	Highest durability (**99.999%**). Max 256,000 IOPS. Supports **Multi-Attach**.
Provisioned IOPS	**io1**	Critical DBs requiring sustained IOPS.	Max 64,000 IOPS. Supports **Multi-Attach**. Legacy compared to io2.

Hard Disk Drives (HDD) - Optimized for large streaming workloads (Throughput)

Cannot be used as a Boot Volume.

Type	Name	Use Case	Key Features
Throughput Optimized	**st1**	Big Data, Data Warehousing, Log processing (Kafka, Splunk).	Max throughput 500 MiB/s. Optimized for sequential I/O.
Cold HDD	**sc1**	Infrequent access, archive data.	**Lowest cost** EBS volume. Max throughput 250 MiB/s.

3. EBS Snapshots

- **Mechanism:** Incremental backups. Only blocks that have changed since the last snapshot are saved.
- **Storage:** Stored in **Amazon S3** (but you cannot access them directly in your S3 buckets).
- **Scope:** Regional. To use a snapshot in another region, you must **Copy** it.
- **Key Features:**
  - **Recycle Bin:** Can retain deleted snapshots for a specified period (1 day to 1 year) to recover from accidental deletions.
  - **Fast Snapshot Restore (FSR):** Eliminates the latency of "initializing" (warming up) a volume created from a snapshot. Expensive.
  - **Data Lifecycle Manager (DLM):** Automates the creation, retention, and deletion of snapshots based on schedules and tags.

4. AMI (Amazon Machine Image)

- **Definition:** A blueprint used to create EC2 instances (contains OS, app server, and applications).
- **Scope: Region-bound**. To launch an AMI in another region, you must first copy the AMI to that region.
- **Sources:**
  - **Public AMIs:** AWS provided (Amazon Linux 2, Ubuntu, etc.).
  - **Your Own AMIs:** Created from your customized instances.
  - **AWS Marketplace:** Paid/Subscription AMIs from vendors (e.g., hardened security images, firewall appliances).
- **Lifecycle:** Create (from instance/snapshot) -> Register -> Copy (cross-region) -> Deregister (delete).

5. EC2 Instance Store ("Ephemeral Storage")

- **Definition:** Physical storage directly attached to the host computer running your instance.
- **Performance:** extremely high I/O (millions of IOPS) and low latency.
- **Volatility (Exam Critical):** It is **ephemeral**.

    - **Data Persists:** On **Reboot**.
    - **Data LOST:** On **Stop**, **Terminate**, or **Hardware Failure**.
- **Use Cases:** Cache, buffers, scratch data, high-performance computing (HPC), temporary content that is replicated elsewhere.
- **Snapshots:** You **cannot** snapshot Instance Store volumes.

6. EBS Multi-Attach

- **Function:** Allows a *single* EBS volume to be attached to *multiple* EC2 instances simultaneously.
- **Constraints:**
    - **Same AZ** only.
    - **io1** and **io2 Block Express** volumes only.
    - Supported on Nitro-based instances.
    - **Filesystem Warning:** You need a **Cluster-aware file system** (e.g., GFS2, OCFS2). Standard filesystems (ext4, XFS, NTFS) will result in data corruption.
- **Use Case:** Clustered Linux applications (e.g., Teradata), high availability updates.

7. EBS Encryption

- **Mechanism:** Uses AWS KMS (Key Management Service) with AES-256.
- **Scope:** When encryption is enabled:
    1. Data at rest inside the volume.
    2. Data in transit between the instance and the volume.
    3. All snapshots created from the volume.
    4. All volumes created from those snapshots.
- **Performance:** Minimal impact on latency.
- **Encryption by Default:** You can enable account-level encryption by default for a specific region.

8. Amazon EFS (Elastic File System)

- **Definition:** Managed NFS (Network File System) that can be mounted on many EC2 instances simultaneously.
- **Protocol:** NFSv4. Works with **Linux** instances only (POSIX compliant).
- **Scope: Multi-AZ**. Data is stored across multiple AZs in a region (High Availability).
- **Scaling:** Automatically scales storage capacity and throughput (Petabyte scale).
- **Storage Classes:**
    - **EFS Standard:** Frequent access, Multi-AZ.
    - **EFS Standard-IA (Infrequent Access):** Lower cost for files not accessed daily.
    - **EFS One Zone:** Stores data in a single AZ (lower cost, lower availability).
- **Performance Modes:**

- **General Purpose:** Web serving, CMS (default).
- **Max I/O:** Big Data, parallel processing (higher latency, higher throughput).

9. Comparison: EBS vs. EFS vs. Instance Store

Feature	EBS (Elastic Block Store)	EFS (Elastic File System)	Instance Store
Type	Block Storage (Network Drive)	File Storage (Network File System)	Block Storage (Physical Disk)
Persistence	Durable (persists across stops)	Durable (persists across stops)	**Ephemeral** (lost on stop)
Access	Single Instance (mostly)	**1000s of Instances** simultaneously	Single Instance only
Availability	Single AZ	**Multi-AZ** (Regional)	Single AZ (Host bound)
Performance	Low Latency	Higher Latency than EBS	**Lowest Latency** (Highest IOPS)
Backup	Snapshots	AWS Backup / EFS Backup	None (Data is temporary)
OS Support	Linux & Windows	**Linux Only**	Linux & Windows